21st National High School Physics Competition Final Problems

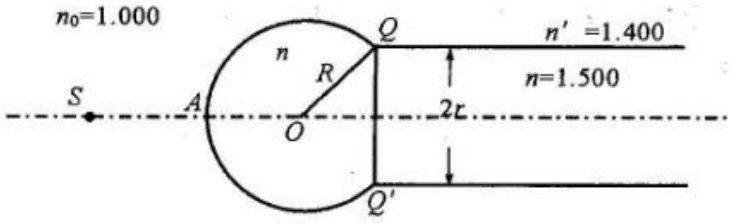
1. (20 points)

Highly transparent fibre core: $n=1.500$, radius r; cladding index $n'=1.400$. Semiconductor laser S, on the axis extension, emits a beam of half-angle 30°. To admit all the light, a spherical cap QAQ′ of radius R, of the core material, is sintered at its end. The figure omits the cladding. S is a point source; the fibre is in air, with index n_0 taken as 1.000.

1. For the whole beam to reach the cap, what condition must distance x from source S to A satisfy?

2. For $R=1.8r$, S–A separation R, axis angle α: for α at $\alpha_1=30°$、$\alpha_2=25°$ and $\alpha_3=20°$, can each of the three rays propagate in the fibre by total internal reflection?

1. Fibre: highly transparent core of index $n = 1.500$, with circular core radius r; cladding index $n' = 1.400$. A semiconductor laser S, on the axis extension, emits a 30° half-angle beam. To admit it all, the fibre end has a sintered cap QAQ' of radius R, of the core material. As shown, cladding is omitted. S is a point source. The fibre is in air of index n_0, taken as 1.000.



1. For all rays to reach the cap, what condition must distance x from source S to A satisfy?

2. For $R = 1.8r$, S - A distance R, let the axis angle be α. For α at $\alpha_1 = 30°$, $\alpha_2 = 25°$, $\alpha_3 = 20°$, can each ray travel in the fibre by total internal reflection?

2. (20 points)

Use relativistic energy and momentum to analyse and justify the following.

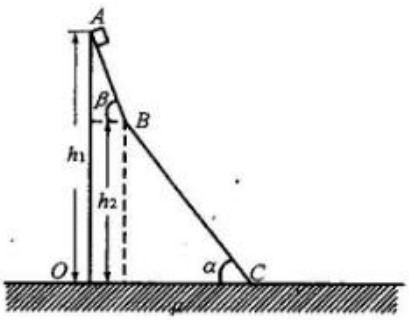
1. A photon colliding with a free electron at rest in vacuum cannot have its energy fully absorbed.

2. A photon on a metal can be fully absorbed, ejecting an electron by the photoelectric effect.

3. (25 points)

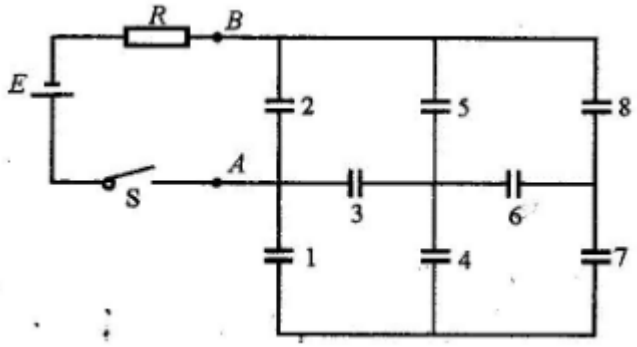
Shown wooden wedge OABC ($M=30.0$kg) rests on level ground; AB has slope $\beta=60°$, and BC slope $\alpha=45°$. A very short smooth arc joins AB, BC at B. Release a point block, $m=4.00$kg, from rest at A, height $h_1=2.80$m above ground. It slides without friction on the wedge. Maximum static and sliding ground friction have coefficients $\mu=6.00\times10^{-2}$. The height of B above the table is $h_2=2.00$m. Ignore both speed changes at B. Find work done on the wedge from A to the bottom C by the sliding block. Take $g=10.0\text{m/s}^2$.

3. Wooden wedge $OABC$ of mass $M = 30.0$ kg rests on level ground. AB has slope $\beta = 60°$, BC has $\alpha = 45°$. AB and BC join at B in a tiny smooth arc. A small block $m = 4.00$ kg, treated as a point. Released from rest at A, $h_1 = 2.80$ m above ground, it slides down the frictionless slope. Maximum static and sliding ground friction coefficients $\mu = 6.00 \times 10^{-2}$. Table height of B: $h_2 = 2.00$ m. Neglect both speed changes at B. Find the work on the wedge as the block slides from A to the base C. Take $g = 10.0\ \mathrm{m \cdot s^{-2}}$.



4. (25 points)

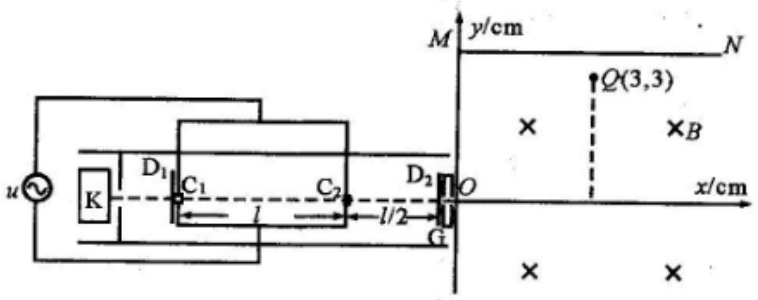
Here E is source emf (negligible internal resistance); R: resistance; S: switch. Right of A、B, 8 identical ideal capacitors, each C. From closing S to full charge, find the heat in R.
In your solution, mark on the diagram the assumed sign of charge on every capacitor plate.



5. (25 points)

K emits positive particles continuously at various speeds, with charge q and mass m, all along the dashed line. D_1、D_2 are timed shutters; C_1、C_2 are gates; C_1 adjoins D_1; gate spacing l. Both gates carry AC voltage $u=U_0\sin(\frac{2\pi}{T}t)$ of period T. Passage occurs only instantaneously near zero voltage. Energy modulator G adjoins D_2, $l/2$ from gate C_2. At time t, G changes energy by $\Delta E_t=E_0\sin(\frac{2\pi}{T}t+\frac{\pi}{4})$, keeping velocity direction. $E_0=\frac{\sqrt{2}}{4}\times m\frac{l^2}{T^2}$. Past G, particles enter at O the uniform field right of G. Its magnitude is B, into the page; all is in vacuum. As shown, origin O defines Cartesian axes Oxy; MN is parallel to the x axis. At $t=0$ open D_1, D_2 together to admit particles to C_1. At $t=3T/4$, close D_1 to stop entry to C_1; at $t=10T/4$, close D_2 to stop entry to G. The fastest entry at O passes Q at (3cm, 3cm). For the slowest entry at O to pass Q, where must boundary MN be shifted parallel to itself in plane Oxy?

5. Source K shown emits particles continuously at various speeds, with positive charge q and mass



m, all along the dashed line. Timed shutters D_1、D_2 open and close. C_1、C_2 are gates; C_1 adjoins D_1, spacing l. Both gates carry AC voltage $u = U_0\sin\left(\frac{2\pi}{T}t\right)$, of period T. Particles pass only for an instant near zero gate voltage. Modulator G adjoins D_2; distance to gate C_2 is $l/2$. At time t, passing G changes the energy by $\Delta E_t = E_0\sin\left(\frac{2\pi}{T}t + \frac{\pi}{4}\right)$; direction unchanged. $E_0 = \frac{\sqrt{2}}{4}\times$

$m\frac{l^2}{T^2}$. Past G, enter at O the field right of G.

Uniform magnitude B, into the page; all is in vacuum. Define the Cartesian axes shown with origin O: Oxy; boundary MN is parallel to the x axis. At $t = 0$, open D_1, D_2 together, admitting to C_1; at $t = 3T/4$, close D_1 to stop entry to C_1; at $t = 10T/4$ close D_2, stopping entry to G. The fastest particle from O passes point Q, at (3 cm,3 cm). **For the slowest particle entering at O also to pass Q, where in plane Oxy must MN be** shifted parallel to itself?

6. Shown: blocks a、b、c、d on a smooth level table,



all of mass m. a、b spring: natural length l, stiffness k_1; c、d are joined by a spring of natural length l, stiffness k_2. Their centres are collinear. b、c collisions are perfectly elastic and brief. Both springs start at natural length. c、d rest; a、b move right at velocity v_0. Discuss their motion quantitatively.

1. $k_1 = k_2$: how do all four move relative to the table?

2. $k_1 = 4k_2$: how do all four move relative to the table?

6. (25 points)

Blocks a、b、c、d, of mass m, lie on a smooth level table. A spring of natural length l joins a、b; its stiffness is k_1. A spring of natural length l, stiffness k_2, joins c、d. All four centres lie on a line. b、c collisions are perfectly elastic and very brief. Both springs initially have natural length; c、d rest, while a、b move right at the same velocity v_0. Discuss quantitatively:

（1） For $k_1=k_2$, how do the four blocks move relative to the table?

（2） For $k_1=4k_2$, how do the four blocks move relative to the table?